

GNSS-Aided Semantic Landmark Particle Filtering for Row-Level Localisation in Vineyards

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Abstract—Reliable localisation in vineyards is hindered by row-level perceptual aliasing: parallel crop rows produce similar LiDAR observations, so geometry-only localisation can settle in an adjacent corridor. We study a Semantic Landmark Particle Filter (SLPF) that combines 2D LiDAR, trunk and support-pole detections, and a noisy GNSS prior. The detections are converted into surveyed semantic walls that impose row-aligned structural constraints in the particle-filter likelihood. An evidence-first evaluation reports raw map-frame APE, aligned APE, RPE, and separate in-row and headland operational metrics over two traversals and three seeds. The results support a bounded conclusion: semantic-wall constraints are a useful component for row-level localisation in the evaluated vineyard, subject to map quality, GNSS availability, and landmark visibility.

I. INTRODUCTION

Long-term autonomy in vineyards requires a global pose estimate that remains meaningful while a robot enters and leaves repetitive crop rows. Parallel rows create perceptual aliasing in LiDAR: several corridors can produce locally consistent range measurements even when only one is physically correct. Seasonal foliage also changes the scene while trunks and support poles remain comparatively stable. GNSS can bound global drift, but its noise and multipath errors are most problematic near headland transitions.

We propose a Semantic Landmark Particle Filter (SLPF) that uses trunk and support-pole detections together with 2D LiDAR and a noisy GNSS prior. The central representation is a semantic wall: neighbouring stable landmarks are grouped into a row-aligned structural constraint rather than treated as unrelated points. These constraints are evaluated in the particle-filter likelihood and complement, rather than replace, geometric and GNSS evidence.

This paper makes three contributions. First, it defines a semantic-wall likelihood for row-level global localisation in a surveyed vineyard map. Second, it makes the evaluation semantics explicit by reporting map-frame and aligned errors together, using in-row frames for row identity, and using headland cross-track and recovery metrics for transitions. Third, it provides a provenance-checked evidence bundle and claim matrix so that the paper tables, report, and manuscript use one frozen baseline and one recorded source.

II. RELATED WORK

Particle-filter localisation is useful in agricultural environments because it can represent non-Gaussian uncertainty caused by wheel slip, imperfect maps, and repetitive structure [1], [2]. Standard AMCL relies primarily on geometric agreement and can therefore preserve an incorrect

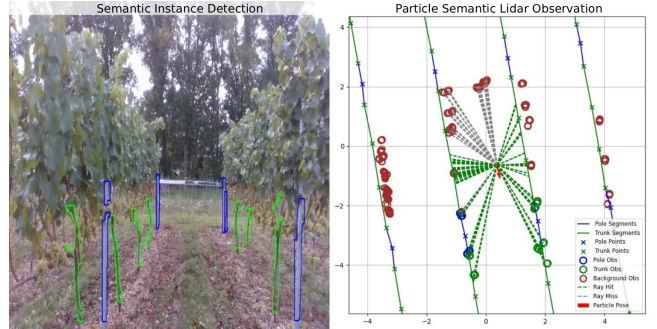


Fig. 1: Example semantic detections projected into the surveyed landmark map. The semantic-wall representation preserves row structure while using sparse trunk and pole observations.

row hypothesis when adjacent corridors look alike. GNSS-aided localisation provides a complementary global cue, but multipath, drift, and outages make a GNSS-only estimate insufficient [3], [4].

Semantic perception has been used to improve agricultural navigation and mapping. Trunk detections, semantic keypoints, and graph-based representations provide more distinctive structure than foliage or raw range data alone [5], [6], [7]. Visual SLAM remains a useful reference for locally smooth motion, but global row identity is not guaranteed by local visual consistency [8]. Our formulation differs by embedding stable landmark structure directly into a probabilistic global localisation likelihood and by separating row identity from headland transition evaluation.

III. SEMANTIC LANDMARK PARTICLE FILTER

A. Semantic-wall map

The fixed map contains row-aligned landmark groups. For row r , the ordered trunk and pole landmarks define a centreline and two offset boundaries. The boundaries form semantic walls that approximate the corridor occupied by the robot. The map is constructed before localisation and is not updated online. A camera detector supplies landmark observations in the robot frame; after projection into the LiDAR frame, observations are associated with the nearest compatible semantic class and row segment.

B. Likelihood and GNSS prior

At time t , each particle $x_t^{(i)}$ is propagated using odometry and scored against the LiDAR scan, semantic observations,

and GNSS measurement. The log weight is

$$\log w_t^{(i)} = \ell_{\text{lidar}}(z_t | x_t^{(i)}) + \lambda_s \ell_{\text{wall}}(s_t | x_t^{(i)}) + \lambda_g \ell_{\text{gnss}}(g_t | x_t^{(i)}). \quad (1)$$

The wall term penalises observations that are inconsistent with the row-aligned boundaries and rewards compatible landmark-to-wall distances. The GNSS term is a robust Huber penalty with a fixed outlier threshold; it is a prior and does not overwrite the particle pose. Semantic penalties are capped to prevent a small number of detector errors from dominating the update.

The normalised weights are used for effective sample size and resampling,

$$N_{\text{eff}} = \frac{1}{\sum_i (w_t^{(i)})^2}. \quad (2)$$

The estimate uses a weighted position mean and circular heading mean, with the maximum-weight particle used when heading dispersion is high. The paper-facing configuration is fixed as `alpha_huber3_cap50`: alpha pose smoothing, Huber threshold 3.0, semantic penalty cap 50, a single row-likelihood update, and no delayed row correction or GNSS row gating.

IV. EXPERIMENTAL PROTOCOL

Experiments use a Thorvald platform equipped with an Intel RealSense D435i RGB-D camera, SICK Tim7 2D LiDAR, and RTK-GNSS. The compact test vineyard contains 10 rows of approximately 16.86 m length and a mean row spacing of approximately 2.50 m. The same surveyed semantic-wall map is used for both traversals. RTK poses provide the quantitative reference; degraded GNSS contains zero-mean noise, correlated drift, and occasional jump-like errors.

The canonical evidence contract requires traversals `rh_run1` and `rh_run2`, seeds `{11,22,33}`, and one row for each method-traversal-seed-profile key. The committed controlled matrix is used as the current evidence source. A prescribed CUDA smoke test reached the runtime but reported that CUDA was unavailable; a short CPU smoke completed, while the full CPU fallback was not completed. Consequently, the compact bundle records the controlled matrix as a fallback source and does not relabel it as a fresh GPU rerun.

A. Metric semantics

Raw APE is computed in the map frame and aligned APE is reported separately; alignment must not hide a global map-frame error. RPE is reported at fixed spatial segments. Cross-track error is the distance to the assigned ground-truth row centreline. Row correctness and wrong-row duration are evaluated only on in-row frames, where row identity is physically meaningful. Headland frames are evaluated with cross-track and transition-recovery metrics because nearest-row identity is ambiguous outside a corridor. Total wrong-row duration is retained only as an all-frame diagnostic.

The detector is treated as an input to localisation. No detector accuracy claim is made because the supplied repository does not contain the required SemanticBLT validation YAML. Similarly, visual-SLAM and GNSS-stress numbers are not invented when their reconciled canonical artifacts are absent.

V. RESULTS

Table I records the reproducible reference point for the paper-facing SLPF configuration. Across the two traversals, the controlled source gives raw APE means of 1.01 m and 0.95 m, and aligned APE means of 0.97 m and 0.88 m, respectively. The separation is important: aligned error describes trajectory consistency after alignment, whereas raw error retains map-frame displacement. RPE is reported alongside both APE forms rather than used as a substitute for global accuracy.

Table II shows why the row metrics are split by context. In-row cross-track means are 1.04 m and 1.18 m, with in-row row-correct fractions of 0.72 and 0.64. Headland cross-track means are reported independently. The bundle also retains headland recovery distance and all-frame wrong-row duration for diagnosis, but the manuscript does not interpret a nearest-row label in a headland as a unique corridor identity.

A. Evidence boundaries and failure modes

The evidence supports the use of semantic walls as a row-structured likelihood term in the evaluated map. It does not establish performance across vineyards, seasons, or unseen maps. The static map assumes that row geometry and stable landmarks remain sufficiently consistent between survey and deployment. Extended GNSS bias together with landmark occlusion can leave the particle set in an adjacent corridor until informative observations reappear.

The canonical bundle currently contains no reconciled RTAB-Map or GNSS-degradation metric files, so those comparisons remain protocol requirements rather than paper claims. The same boundary applies to detector accuracy. The retained qualitative figures may motivate future comparisons, but they do not replace source-backed tables.

VI. LIMITATIONS

The evaluation covers one compact vineyard, one fixed semantic-wall map, and two traversals. It therefore tests a bounded operational question rather than universal row disambiguation. Performance depends on stable map geometry, GNSS that remains sufficiently informative during global transitions, and visible trunk or pole detections. The method is not evaluated as a complete visual-GNSS fusion system, and no claim is made about closed-loop control accuracy. A future release should add reconciled GNSS-stress, visual-SLAM, runtime, and detector-validation artifacts before expanding the claims.

TABLE I: Canonical SLPF evidence from the frozen baseline source. Values are mean $\pm$ seed standard deviation over seeds 11, 22, and 33. Raw APE remains map-frame; aligned APE is a separate diagnostic.

Method	Traversal	Raw APE (m)	Aligned APE (m)	RPE 2m (m)
baseline_alpha_huber3_cap50	rh_run1	1.01 ± 0.14	0.97 ± 0.13	3.37 ± 0.07
baseline_alpha_huber3_cap50	rh_run2	0.95 ± 0.09	0.88 ± 0.04	3.38 ± 0.05

TABLE II: Operational metrics with row identity evaluated in-row and transition behaviour evaluated on headland frames. Wrong-row duration is not used to label headland row identity.

Method	Traversal	In-row XT (m)	In-row correct	In-row wrong (s)	Headland XT (m)	Headland recovery (m)
baseline_alpha_huber3_cap50	rh_run1	1.04 ± 0.07	0.72 ± 0.02	214.67 ± 15.43	1.58 ± 0.01	1.84 ± 0.24
baseline_alpha_huber3_cap50	rh_run2	1.18 ± 0.05	0.64 ± 0.01	253.33 ± 4.99	1.69 ± 0.06	1.78 ± 0.07

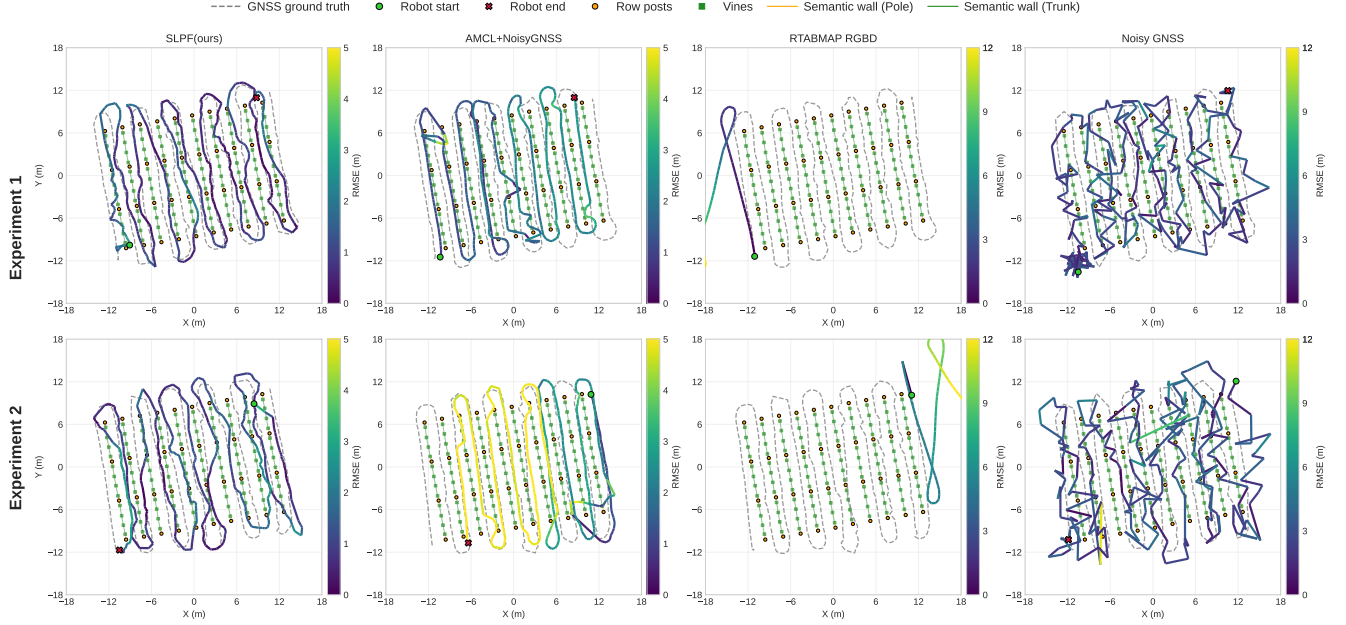


Fig. 2: Recorded trajectory comparisons retained as qualitative context. The canonical numeric claims are generated from the evidence bundle; visual overlays are not used to infer unreported metrics.

VII. CONCLUSION

We presented an evidence-first evaluation of a GNSS-aided Semantic Landmark Particle Filter for row-level localisation in repetitive vineyards. Semantic walls turn sparse trunk and pole detections into row-aligned constraints, while robust GNSS fusion provides a complementary global prior. The frozen controlled evidence reports raw and aligned map-frame errors together with in-row row metrics and separate headland transition metrics. The resulting claim is deliberately bounded to the surveyed vineyard and recorded sensing conditions; map quality, GNSS availability, and landmark visibility remain essential assumptions.

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